Answers and talk notes

Equivalent Ratios

What your student is learning

Tonight we scale a recipe up while keeping the ratio the same — and we find the scale factor that makes it work.

Answers

Ask your student to explain before you look.

1. **B.** 4. 3 × 4 = 12, so the scale factor is 4.
2. **A.** 12. The scale factor is 6, so the water is 2 × 6 = 12 cups.
3. **C.** 20. The scale factor is 8 ÷ 2 = 4, so the strawberries are 5 × 4 = 20.
4. **D.** Multiply every amount by 3. Multiply every amount by 3 keeps the same ratio, so the taste stays the same.
5. **C.** 12. 6 × 4 = 24, so the lemons are 3 × 4 = 12.
6. **B.** 3:4. 18 ÷ 6 = 3 and 24 ÷ 6 = 4, so the ratio is 3:4.

Talk about it

What is the scale factor, and how did you use it on the second ingredient?

Home mission

Use a pitcher and a cup. Pretend a drink needs 3 cups of water for 2 cups of juice. Ask your student: how much juice for 9 cups of water? What number did you multiply by?

When they are stuck, ask

- What number do we multiply the first term by to reach the new amount?
- Did the other term get multiplied by that same number?
- Does the new ratio simplify back to the old one?

Try not to say

- Add the scale factor to both numbers.
- Only the ingredient the question asks about has to change.
- Close enough is fine for a recipe.